

# Security, Compliance, and Governance for AI Solutions

## Shared Responsibility Model

A major concept in AWS AI security is the **AWS Shared Responsibility Model**.

([Amazon Web Services, Inc.](https://aws.amazon.com/compliance/shared-responsibility-model/))

AWS is responsible for:

- Physical infrastructure
- Networking
- Hardware
- Managed cloud services

Customers are responsible for:

- Data protection
- IAM permissions
- Model access
- Encryption
- Application security
- Compliance configuration

Example:

- AWS secures Amazon Bedrock infrastructure.
- You secure prompts, datasets, users, and permissions.

## Core Security Principles for AI Systems

The lesson emphasizes several AI security areas:

### Identity and Access Management (IAM)

Use:

- Least privilege access
- Role-based permissions
- Multi-factor authentication (MFA)

Goal:

Only authorized users and services can access AI resources.

### Data Protection

Protect:

- Training data

- Customer prompts
- Generated outputs
- Sensitive information

Methods:

- Encryption at rest
- Encryption in transit
- Key management with AWS KMS
- Data masking/tokenization

## **Network Security**

Use:

- VPCs
- Security groups
- Private endpoints
- API protection

This prevents unauthorized network access.

## **Monitoring and Logging**

AWS services used:

- CloudTrail
- CloudWatch
- GuardDuty
- Security Hub

Purpose:

- Detect threats
- Monitor AI activity
- Audit compliance
- Track user actions

## **Compliance in AI Solutions**

AI systems may need to comply with regulations and standards such as:

- GDPR
- HIPAA
- ISO standards
- SOC reports
- Industry-specific regulation

AWS provides compliance-ready infrastructure and certifications. ([Amazon Web Services, Inc.](#))

Key compliance goals:

- Data privacy
- Auditability
- Risk management
- Responsible data usage

## AI Governance

Governance means creating policies and controls for how AI is developed and used.

Important governance practices:

- Human oversight
- Approval workflows
- Risk assessments
- Model versioning
- Access policies
- Usage monitoring

AWS recommends a **platform-centric governance model** where security guardrails are applied consistently across all AI applications. ([AWS Documentation](#))

## Responsible AI

Responsible AI focuses on making AI:

- Fair
- Transparent
- Safe
- Explainable
- Accountable

Risks discussed:

- Bias
- Harmful outputs
- Hallucinations
- Privacy leakage
- Prompt injection attacks

AWS recommends

- Human review
- Content filtering
- Testing and validation
- Monitoring outputs

- Using guardrails

## Generative AI Security Risks

Generative AI introduces new risks beyond traditional software security. ([AWS Documentation](#))

Examples:

- Prompt injection
- Data leakage
- Model manipulation
- Unauthorized model access
- Toxic or unsafe responses

Mitigation strategies

- Input validation
- Guardrails
- Output filtering
- Secure APIs
- Monitoring
- Isolation of sensitive workloads

## AWS Services Commonly Mentioned

Likely AWS services in this lesson:

- Amazon Web Services
- Amazon Bedrock
- Amazon SageMaker
- AWS IAM
- AWS CloudTrail
- Amazon GuardDuty
- AWS KMS
- Amazon CloudWatch